Contributing

How changes are made in this repository. It is a small team on a fixed hackathon schedule, so the rules are short and enforced by CI.

Setup

```
python3 -m venv .venv
source .venv/bin/activate
pip install -e ".[dev]"
```

Branches

- `main` holds the integrated project. Do not push to it directly; open a pull request.
- Work on a short-lived branch named for its scope, for example `capture/tcp-reassembly` or `protocol/field-types`. Delete it after the merge.

Commits

Conventional Commits, English, subject at most 72 characters, imperative mood.

```
feat(capture): implement TCP stream reassembly
test(protocol): cover incomplete message framing
fix(ui): keep hex selection after a theme switch
docs(readme): note the portable project layout
```

The scope names the area (`capture`, `protocol`, `hypothesis`, `ui`, `project`, `report`, `docs`, `ci`). One logical change per commit.

Before pushing

Run the gates locally; CI runs the same ones.

```
QT_QPA_PLATFORM=offscreen pytest tests/ -q
ruff check .
```

CI also audits dependencies (`pip-audit` under `.[dev]`). `ruff format --check` and `mypy .` are deferred for the modules outside this change, as recorded in `docs/AUDIT.md`; run them on the files you touch.

- All tests pass. A new behaviour comes with a test over the real bytes and the real engine, not over a mock.
- No `TODO`, `FIXME`, commented-out code, or debug output.
- No secrets, no personal data, no large binaries outside `presentation/`.
- Do not commit `.venv/`, `__pycache__/`, `*.pcap` outside `tests/fixtures/`, a local `project.madrigal/`, or `.env`. The `.gitignore` covers all of these.

Pull requests

- Keep the title in the commit style above.
- State what changed, why, and how it was checked (the command you ran).
- Do not merge your own pull request; it is reviewed first.

Ownership by area

Area	Directory
Capture engine	`src/capture/`
Protocol and hypothesis engines	`src/protocol/`, `src/hypothesis/`
Window, project, report, presentation	`src/ui/`, `src/project/`, `src/report/`, `presentation/`
Shared contracts and schemas	`docs/CONTRACT.md`, `docs/schemas/` — change by agreement, never silently